# Project Design Phase-II

# Solution Requirements (Functional & Non-functional)

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| Project Name | Strategic product placement analysis: unveiling sales impact with tableau visualization |

**Functional Requirements:**

Following are the functional requirements of the proposed solution.

| FR No | Functional Requirement (Epic) | Sub Requirement (Story / Sub-Task) |
|---|---|---|
| FR-1 | Data Collection and Extraction of Data | ➔ Collect the dataset<br>➔ Understanding the Data<br>➔ Connecting the dataset with Tableau |
| FR-2 | Data Preparation | ➔ Prepare the Data for Visualization |
| FR-3 | Data Visualization | ➔ No of Unique Visualizations |
| FR-4 | Dashboard | ➔ Responsive and Design of Dashboard |
| FR-5 | Story | ➔ No of Scenes of Story |
| FR-6 | Performance Testing | ➔ Utilization of Filters<br>➔ No of Calculation Fields<br>➔ No of Visualizations/ Graphs |

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| FR-7 | Web integration | ➔ Dashboard and Story embed with UI With Flask |
| FR-8 | Project Demonstration & Documentation | ➔ Record an explanation Video for the project end to end solution<br>➔ Project Documentation-Step by step project development procedure |

**Non-functional Requirements:**

Following are the non-functional requirements of the proposed solution.

| **FR No** | **Non-Functional Requirement** | **Description** |
|---|---|---|
| NFR-1 | Usability | The dashboard must feature an intuitive UX, allowing category managers to filter by region, SKU, and shelf position with zero training. It should follow visual best practices (e.g., |
| | | color-blind friendly palettes). |

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| NFR-2 | Security | Data access must be governed by Row-Level Security (RLS), ensuring regional managers only see data for their specific stores. Connection to the data source must be encrypted via SSL/TLS. |
| NFR-3 | Reliability | The system must maintain 99.9% data accuracy between the source ERP/POS system and the Tableau visualization.<br>Automated data quality checks should flag anomalies in sales impact metrics. |
| NFR-4 | Performance | Dashboards must load within 3-5 seconds. Data extracts or Live queries must be optimized (using indexing or partitioning) to ensure |
| | | smooth interaction when toggling between "Before" and "After" placement views. |

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| NFR-5 | Availability | The visualization must be accessible 24/7 via Tableau Server/Cloud, with scheduled maintenance occurring only during off-peak hours to ensure stakeholders can run reports whenever needed. |
| NFR-6 | Scalability | The solution must support a 50% increase in data volume (new store locations or products) and an increase in concurrent users without a degradation in rendering speed. |